

Experiment.7

Aim:- To write metadata of your E-commerce PWA in a Web app manifest file to enable the “add to home screen feature”.

Theory-

❖ What is a Progressive Web App (PWA)?

A **Progressive Web App (PWA)** is a type of web application that leverages modern web technologies to provide a user experience similar to a native mobile application. PWAs combine the benefits of both web and mobile apps, offering features such as offline access, push notifications, and installation on a user's home screen.

Difference Between Regular Web App and PWA

A regular web app is a website designed to be accessible on mobile and desktop devices through a browser. However, a regular web app relies on the browser for functionality and does not support offline access or installation. In contrast, PWAs enhance the user experience by combining the best aspects of web and native apps. PWAs are installable, work offline, load quickly, and offer a seamless app-like experience.

❖ Advantages of PWA

1. **Faster Loading:** PWAs cache assets and serve them quickly, even under poor network conditions.
2. **Offline Support:** Content remains accessible even without an internet connection using service workers.
3. **App-like Experience:** PWAs run in a standalone window, offering a full-screen experience.
4. **Easy Installation:** Users can add the app to their home screen without needing to download it from an app store.
5. **Push Notifications:** Allows engagement with users even when the app is not active.
6. **Real-time Updates:** PWAs automatically update in the background, ensuring users always access the latest version.

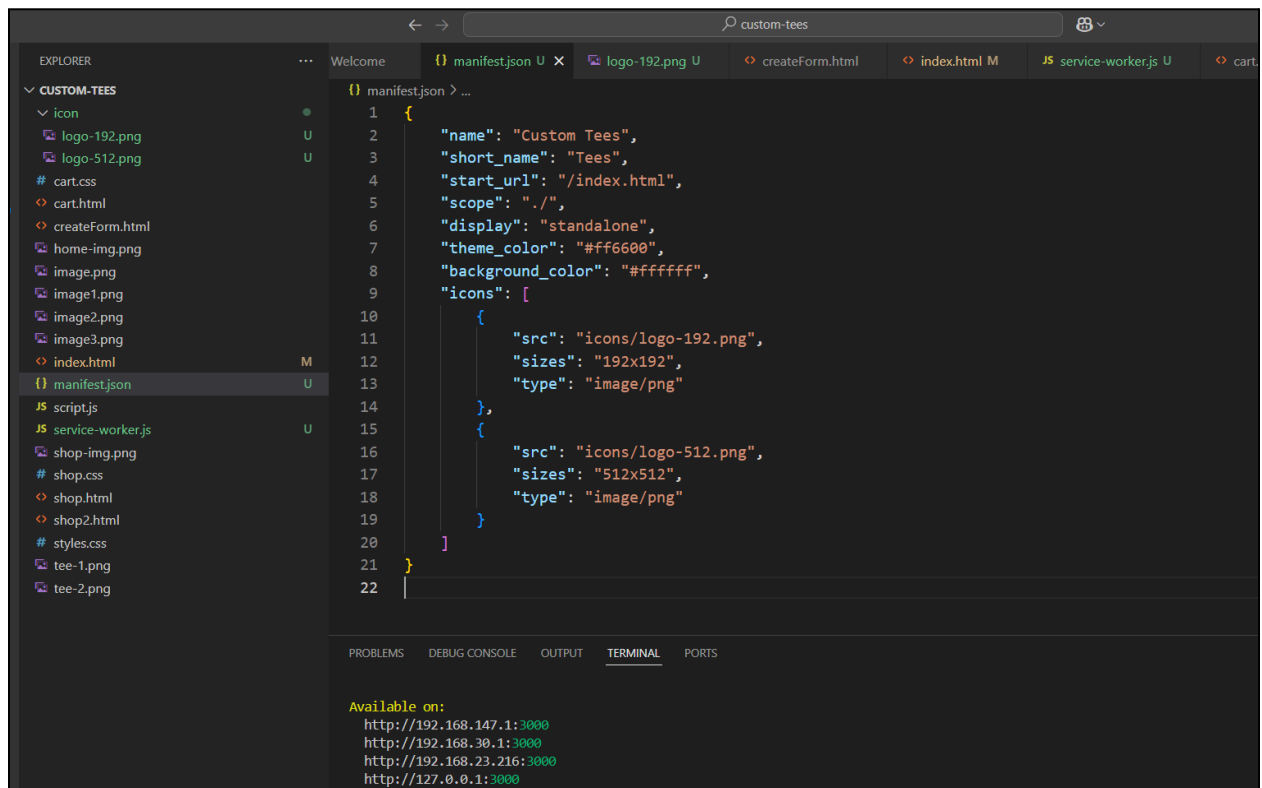
❖ What is a Web App Manifest File?

A **Web App Manifest** is a simple JSON file that defines metadata about the PWA. It includes information such as the app name, icons, start URL, theme color, background color, and display mode. The manifest file allows the app to be installed on the user's home screen and provides a consistent look and feel across different platforms.

How the "Add to Home Screen" Feature Works

The "Add to Home Screen" feature allows users to install the PWA on their device directly from the browser. When the user visits a PWA-compliant website, the browser prompts the user to install the app. Once installed, the app appears on the home screen like a native app and can be launched without opening the browser.

❖ Output:

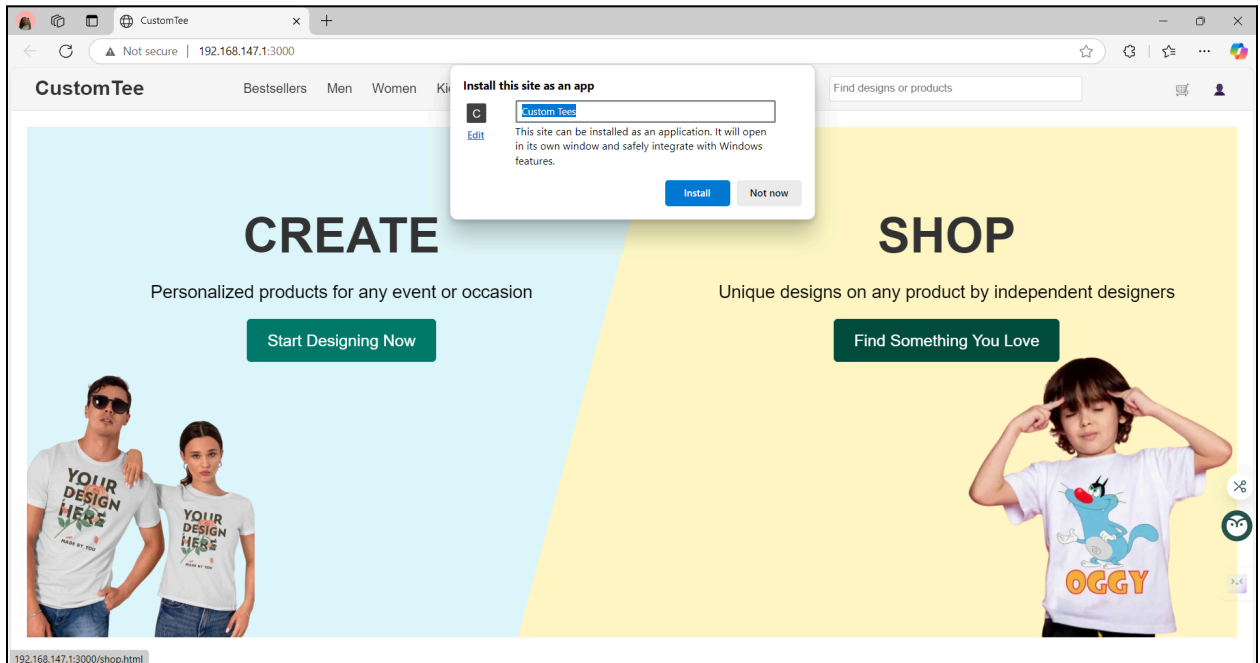
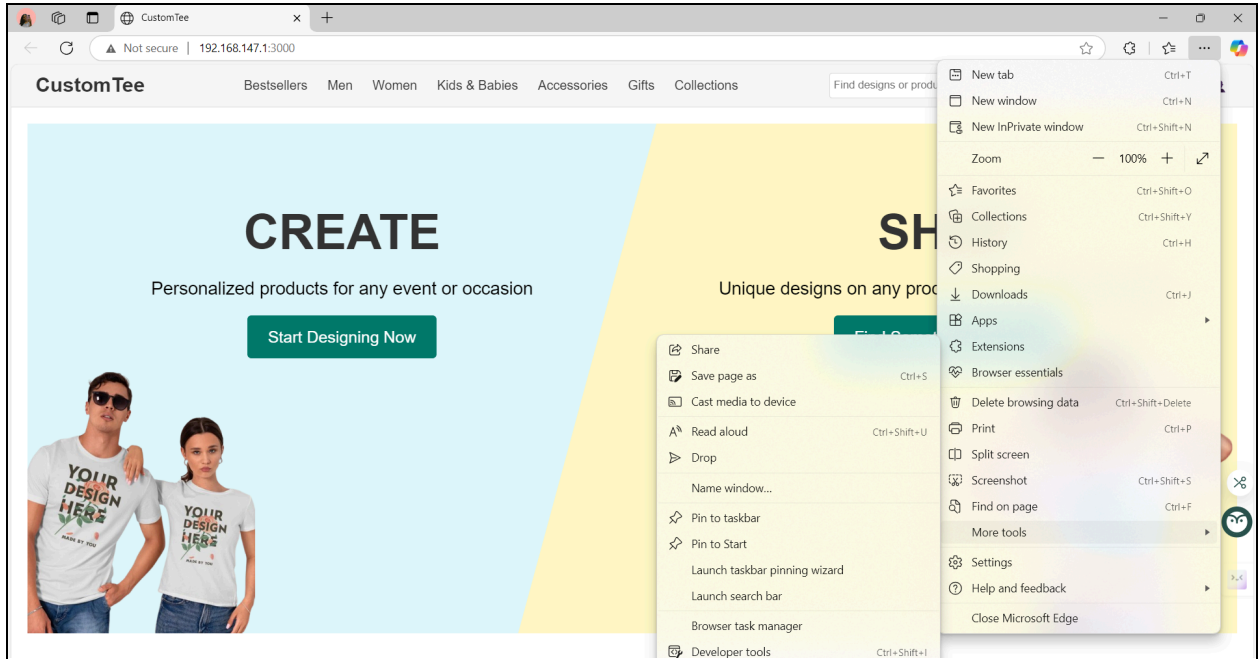


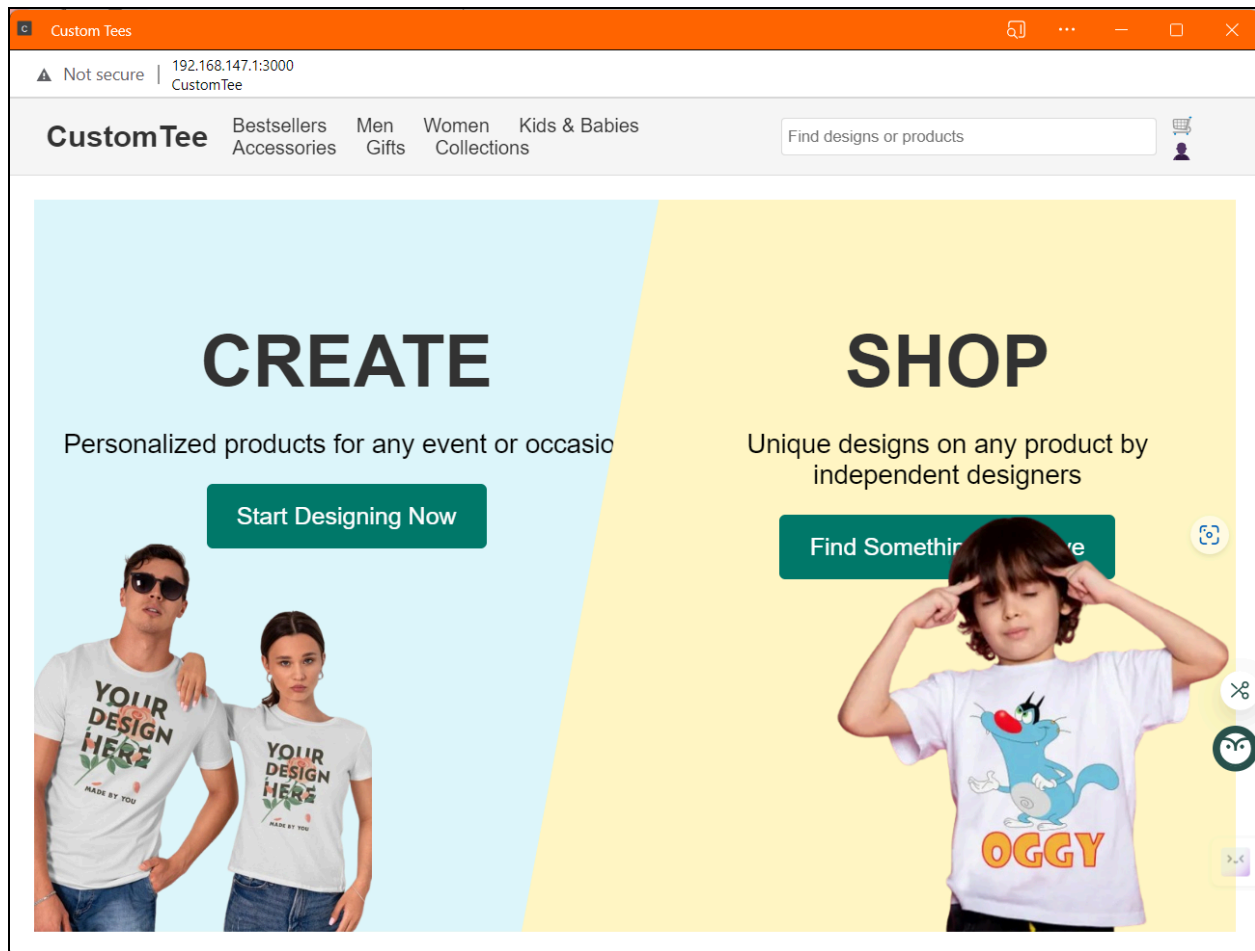
The screenshot shows a web browser window with the address bar displaying "custom-tees". Below the browser, a code editor displays the content of the "manifest.json" file. The file is a JSON object with the following properties:

```
{
  "name": "Custom Tees",
  "short_name": "Tees",
  "start_url": "/index.html",
  "scope": "./",
  "display": "standalone",
  "theme_color": "#ff6600",
  "background_color": "#ffffff",
  "icons": [
    {
      "src": "icons/logo-192.png",
      "sizes": "192x192",
      "type": "image/png"
    },
    {
      "src": "icons/logo-512.png",
      "sizes": "512x512",
      "type": "image/png"
    }
  ]
}
```

At the bottom of the code editor, the terminal output shows the following text:

```
Available on:
http://192.168.147.1:3000
http://192.168.30.1:3000
http://192.168.23.216:3000
http://127.0.0.1:3000
```





❖ Conclusion

Hence, we successfully created metadata for an E-commerce Progressive Web App (PWA) using a Web App Manifest file. The manifest file defines the app's name, icons, start URL, and display options, enabling the "Add to Home Screen" feature. Once installed, the PWA behaves like a native mobile application, providing a faster and more engaging user experience. PWAs offer benefits like offline access, push notifications, and real-time updates, making them a powerful alternative to traditional web and mobile apps.